

Unit-2.

Need for biasing

* Biasing of the bipolar junction transistor is the process of applying external voltage to it.

* In order to use the BJT for any application like amplification, the two junction of the transistor CB and BE should be properly biased according to the required application.

* Depending on whether the two junction of the transistor are forward or reverse bias, a transistor is capable of operating in three different modes.

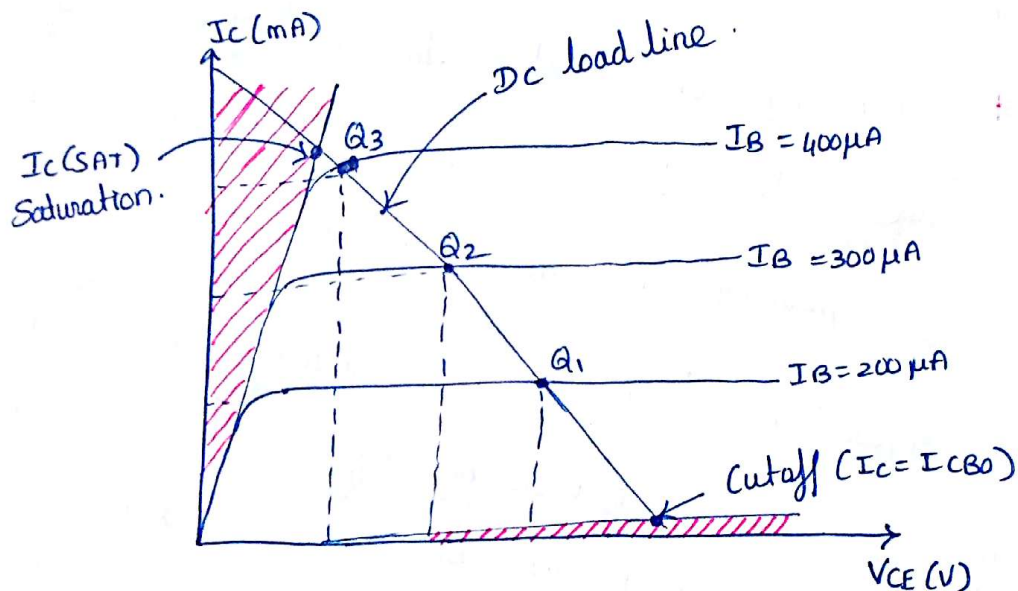
Region of operation	Emitter base junction	collector base junction
* Cut off.	Reverse biased.	Reverse biased
* Active	Forward biased	Reverse biased
* Saturation.	Forward biased.	Forward biased.

* In transistor amplifier circuit, output signal power is always greater than input signal power. Now how this amplification of power is achieved. The DC source supplies the power to the transistor circuit to get the output signal power greater than input signal power.

Operating Point and Stability factor.

* For a transistor circuit to amplify it must be properly biased with dc voltages. The dc operating point between saturation and cutoff is called Q. Point.

* The goal is to set the Q-point such that it does not go into saturation or cutoff when a ac signal is applied.



* The other name for operating conditions are dc operating or quiescent point.

* The Dc load line must be stable for proper operation of transistor. However it shifts with changes in transistor parameter such as β , I_{CO} and V_{BE} . (temperature dependent)

The operating point also varied with changes in temperature. The stability factor indicates the degree of change in operating point due to variation in β , I_{CO} and V_{BE} .

Mode of operation.

Cut off mode.

* The BJT is fully off in this state. In the cutoff mode both the base emitter as well as collector base junction is reverse biased.

Saturation Mode:

* The transistor is fully on in this state. The CB as well as BE junction are forward biased. The BJT operates like a closed switch in the saturation mode.

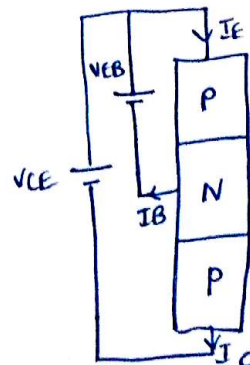
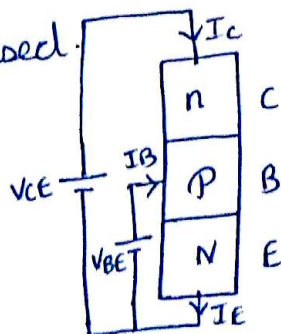
* If a BJT is in saturation mode then it should satisfy the following condition.

$$|I_B| \geq \left(\frac{|I_C|}{\beta_{DC}} \right)$$

* Where β_{DC} is common emitter current amplification factor or current gain.

Active mode:

* In order to use the transistor as an amplifier, it must be operated in the active mode. The BE junction is forward biased whereas CB junction is reverse biased.



Biasing circuit of BJT

* To make the Q point stable different biasing circuits are tried. The Q point is also called as operating bias point, is the point on the DC load line which represents the DC current through the transistor and voltage across it when no AC signal is applied.

* The Q point represents the DC biasing condition. When the BJT is biased such that the Q point is halfway between cutoff and saturation.

* The three circuits or biasing arrangements which are practically used are given below.

- Fixed bias
- Emitter stabilized bias.
- Voltage divider bias.

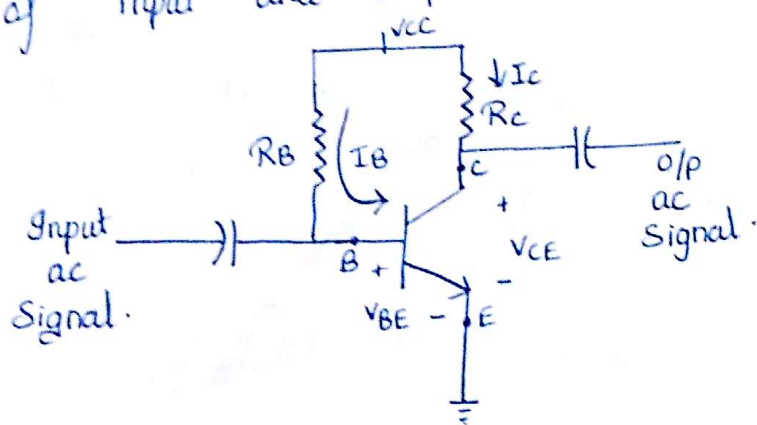
Fixed Bias Circuit

* The fixed bias circuit provides a relatively straightforward and simple introduction to transistor dc bias analysis.

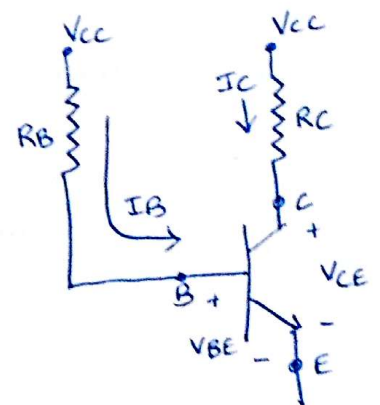
* Even when the network employs an npn transistor instead of pnp transistor, the equation and calculation apply equal to pnp by changing all current direction and voltage polarities.

* For the dc analysis the network can be isolated from the indicated ac levels by replacing the capacitor with an open circuit equivalent.

* In addition the dc supply V_{CC} can be separated of input and output circuits.



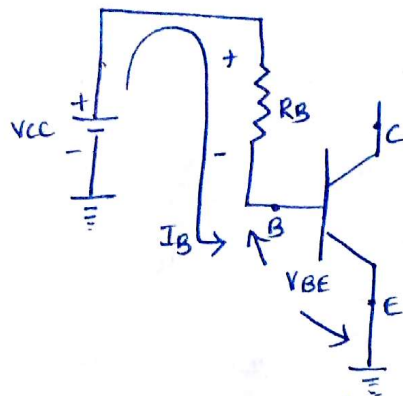
Fixed bias Circuit.



DC equivalent circuit.

Forward Bias of Base emitter

* Consider first the base emitter circuit loop of figure. Apply Kirchhoff's voltage equation in the clockwise direction for the loop.



Base emitter loop.

$$V_{CC} = I_B R_B + V_{BE} \rightarrow (1)$$

$$V_{CC} - I_B R_B - V_{BE} = 0 \rightarrow (2)$$

* Polarity of Voltage drop across R_B established by the indicated direction I_B . Solving the equation for current I_B

$$I_B = \frac{V_{CC} - V_{BE}}{R_B} \rightarrow (3)$$

Collector emitter loop.

* The collector emitter section of network appears in figure. with the indicated direction of current I_C and resulting polarity across R_C

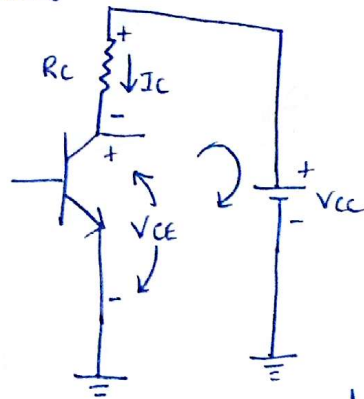
* The magnitude of collector current is related directly to I_B .

$$I_C = \beta I_B$$

* It is note that the base current is controlled by the level of R_B and I_C is related to I_B by a constant β

* change of R_C to any level and it will not affect the level of I_B or I_C

* Applying Kirchhoff's voltage law in clock wise direction around the indicated closed loop.



Collector Emitter loop

$$V_{CE} + I_C R_C = V_{CC} \rightarrow (4)$$

$$V_{CE} = V_{CC} - I_C R_C \rightarrow (5)$$

The voltage across the collector emitter region of transistor, in the fixed bias configuration is the supply voltage less than the drop across R_C

$$V_{CE} = V_C - V_E$$

Where V_{CE} is the voltage from collector to emitter. and V_C & V_E are voltage from collector and emitter.

$$V_E = 0V$$

$$\boxed{V_{CE} = V_C}$$

In addition

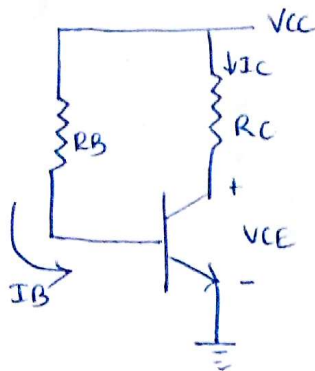
$$V_{BE} = V_B - V_E$$

$$\boxed{V_{BE} = V_B}$$

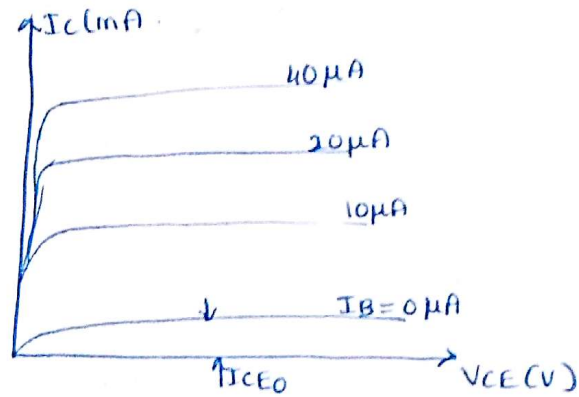
Load Line Analysis.

* The analysis thus far has been performed using of level of I_B corresponding with the resulting Q point

* We will now investigate how the network parameters define the possible range of Q points.



(a) Network



(b) Device characteristics.

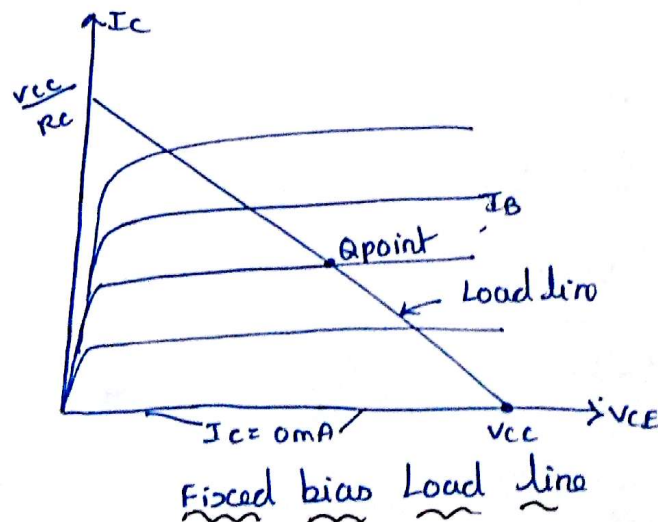
$$V_{CE} = V_{CC} - I_C R_C$$

* The output characteristics of transistor also related the same two variable I_C & V_{CE}

* If we choose I_C to be 0 mA, we are specifying the horizontal axis as the line on which the Q point is located.

$$V_{CE} = V_{CC} - (0) R_C$$

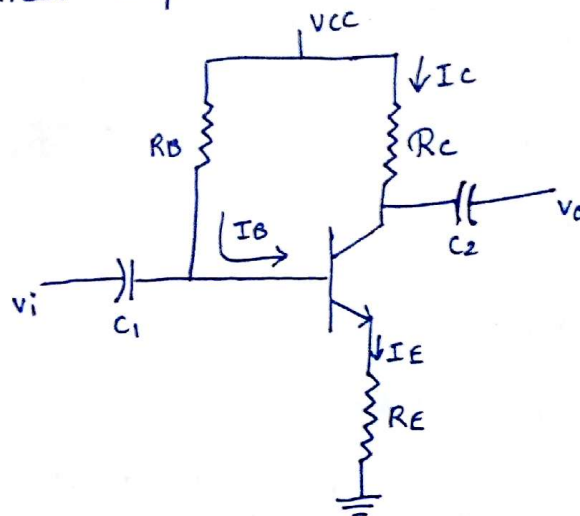
$$V_{CE} = V_{CC}$$



Emitter Stabilized Bias Circuit.

* The DC bias network of figure contains an emitter resistor to improve the stability level over that of fixed bias configuration.

* The analysis will be performed by first examining the base-emitter loop and then using the results to investigate the collector-emitter loop.



Base emitter loop.

* The base emitter loop of the network can be redrawn as shown in figure.

* Applying KVL around the indicated loop in clockwise direction

$$V_{CC} - I_B R_B - V_{BE} - I_E R_E = 0 \rightarrow (1)$$

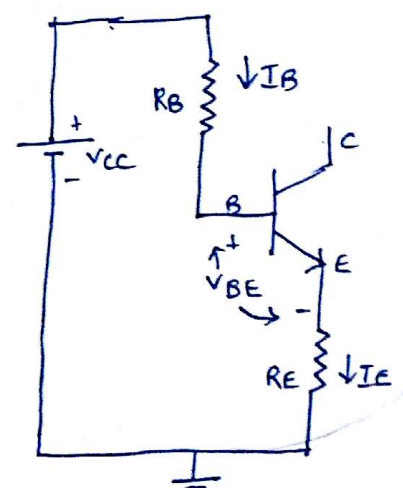
We know that

$$I_E = I_C + I_B \rightarrow (2)$$

* The magnitude of I_C is equal to I_B . so above equation can be written as

$$I_E = \beta I_B + I_B$$

$$I_E = (\beta + 1) I_B \rightarrow (3)$$



* Substituting for I_E in equation ①

$$V_{CC} - I_B R_B - V_{BE} - (\beta + 1) I_B R_E = 0 \rightarrow \textcircled{4}$$

* Grouping terms will then provide the following

$$-I_B (R_B + (\beta + 1) R_E) + V_{CC} - V_{BE} = 0$$

* Multiplying through by (-1) we have.

$$I_B (R_B + (\beta + 1) R_E) - V_{CC} + V_{BE} = 0$$

$$I_B (R_B + (\beta + 1) R_E) = V_{CC} - V_{BE} \rightarrow \textcircled{5}$$

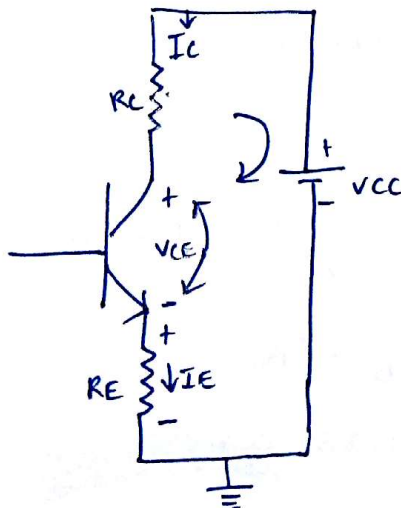
* For solving I_B

$$I_B = \frac{V_{CC} - V_{BE}}{(R_B + (\beta + 1) R_E)} \rightarrow \textcircled{6}$$

* Note that the only difference between this equation for I_B and that obtained for the fixed bias configuration is the term $(\beta + 1) R_E$

Collector - Emitter loop.

* The collector emitter loop is redrawn as shown in figure.



* Writing KVL for the indicated loop in clockwise direction.

$$I_E R_E + V_{CE} + I_C R_C - V_{CC} = 0 \rightarrow (7)$$

* Substituting $I_E \equiv I_C$ and grouping terms gives.

$$V_{CE} - V_{CC} + I_C (R_C + R_E) = 0 \rightarrow (8)$$

and

$$V_{CE} = V_{CC} - I_C (R_C + R_E)$$

* The single subscript voltage V_E is the voltage from emitter to ground and determined by

$$V_E = I_E R_E \rightarrow (9)$$

* While the voltage from collector to ground can be determined from

$$V_C = V_{CE} + V_E \rightarrow (10)$$

$$V_C = V_{CE} + V_E$$

$$V_C = V_{CC} - I_C R_C \rightarrow (11)$$

* The voltage at the base with respect to ground can be determined from

$$V_B = V_{CC} - I_B R_B \rightarrow (12)$$

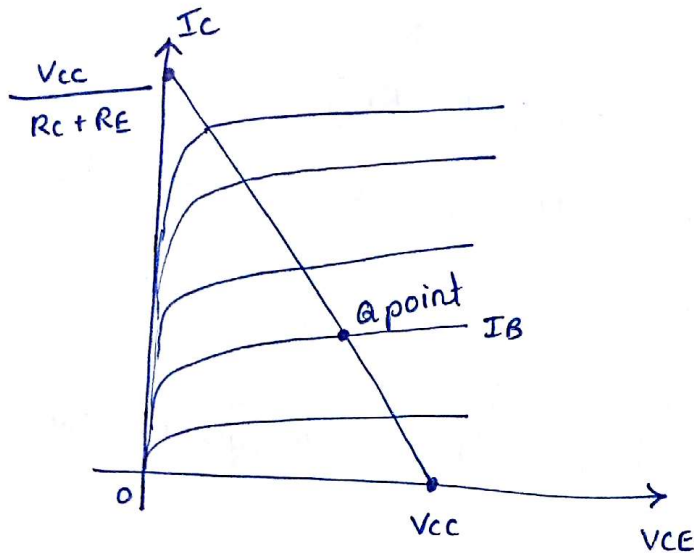
$$V_B = V_{BE} + V_E \rightarrow (13)$$

Improved bias stability

* The addition of emitter resistor to the dc bias of BJT provides improved stability, that is, the dc bias current and voltage remain closer to where they set by circuit when outside conditions, such as temperature, transistor beta change.

Load Line Analysis.

* The load line analysis of the emitter bias network is only slightly different from that encountered for the fixed bias configuration.



* The collector emitter loop equation that defines the load line

$$V_{CE} = V_{CC} - I_C (R_C + R_E)$$

→ choosing $I_C = 0 \text{ mA}$

$$V_{CE} = V_{CC} \big|_{I_C = 0 \text{ mA}} \longrightarrow (14)$$

→ choosing $V_{CE} = 0 \text{ V}$

$$I_C = \frac{V_{CC}}{R_C + R_E} \longrightarrow (15)$$

Voltage Divider Bias

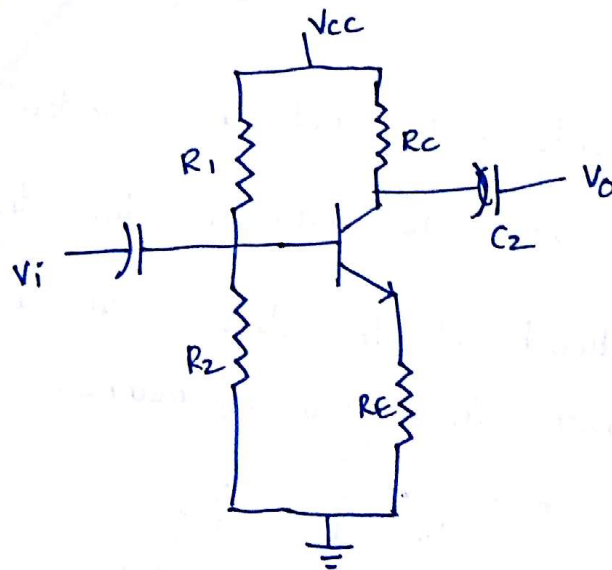
In Fixed bias and emitter stabilized bias configuration the bias current I_C and V_{CE} were a function of current gain (β) of transistor.

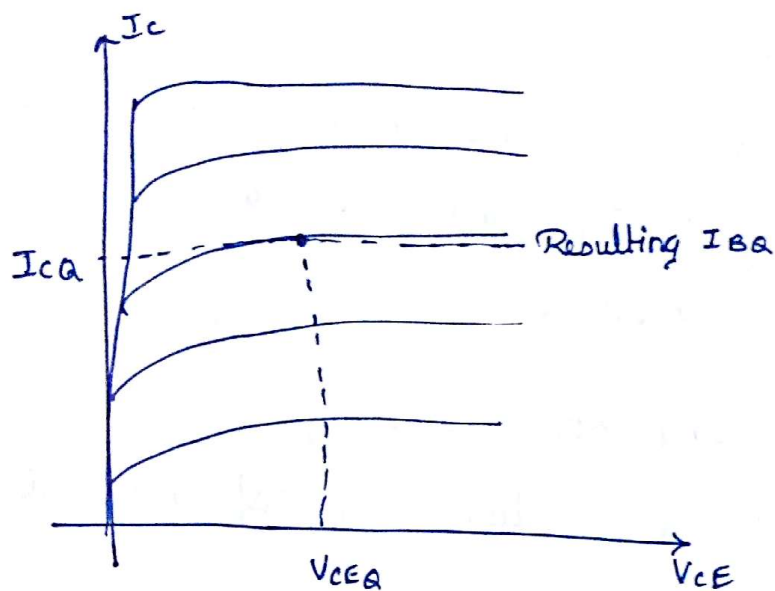
β is temperature sensitive, especially for silicon transistor. and the actual value of β is usually not well defined.

If analyzed on a exact bias the sensitivity to change in β is quite small.

If the circuit parameters are properly chosen. the resulting levels of I_C & V_{CE} can totally independent of β .

The voltage divider bias configuration is shown in figure.





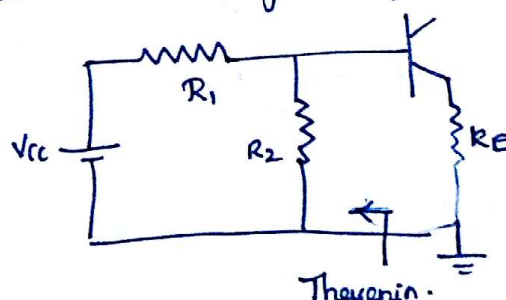
The level of I_{BQ} will change with the change in β but the operating point on the characteristics defined by I_{CQ} and V_{CEQ} can be remain fixed.

There are two methods than can be applied to analyze the voltage divider configuration.

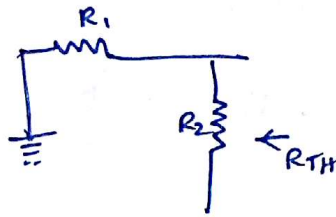
- 1) Exact method. (any voltage divider bias)
- 2) Approximate method. (specific conditions are satisfied)

Exact Analysis

The input side of the network can be drawn as shown in figure for dc analysis. The thevenin equivalent network for the network to the left of the base terminal can then be found in following manner.

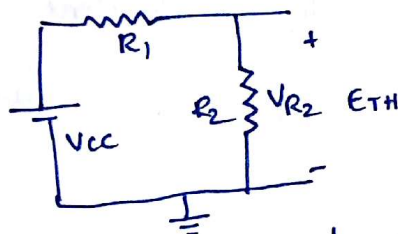


R_{TH} : The voltage source is replaced by short circuit equivalent as shown in figure



$$R_{TH} = R_1 \parallel R_2$$

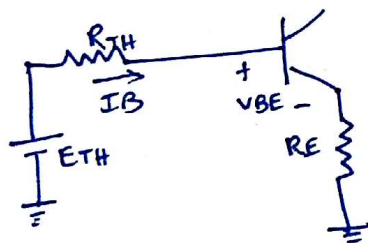
E_{TH} : The voltage source V_{CC} is returned to the network and the open circuit



Applying the Voltage divider rule.

$$E_{TH} = V_{R2} = \frac{R_2 V_{CC}}{R_1 + R_2}$$

The Thevenin network is then redrawn as shown in figure and I_{BQ} can be determined by first applying Kirchhoff's Voltage law in clockwise direction for the loop



$$E_{TH} - I_B R_{TH} - V_{BE} - I_E R_E = 0$$

Substituting $I_E = (1 + \beta) I_B$

$$I_B = \frac{E_{TH} - V_{BE}}{R_{TH} + (\beta + 1) R_E}$$

Once I_B is known, the remaining quantities of the network can be found in the same manner as developed for emitter bias configuration

$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$

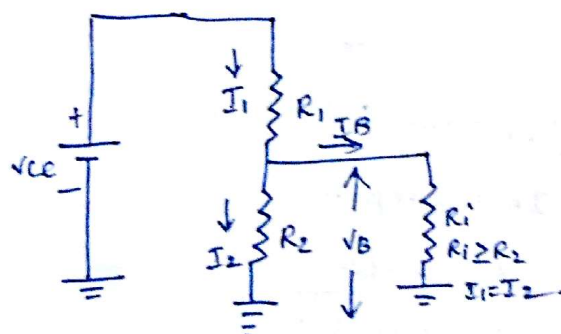
Approximate Analysis

The input section of the voltage divider configuration can be represented by the network.

The Resistance R_i is the equivalent resistance between base and ground for the transistor with an emitter resistor R_E

If R_i is much larger than the resistance R_2 the current I_B will be much smaller than I_2 and I_2 will be approximately equal to I_1

If we accept the approximation that I_B is essentially zero amperes compared to I_1 & I_2 then $I_1 = I_2$ if R_1 & R_2 can be considered series elements



The Voltage across R_2 , which is actually the base Voltage, can be determined using the Voltage Divider rule.

$$V_B = \frac{R_2 V_{CC}}{R_1 + R_2} \rightarrow (1)$$

Since $R_i = \beta R_E$ the condition that will define whether the approximate can be applied will be the following

$$\beta R_E \geq 10 R_2 \rightarrow (2)$$

Once V_B is determined the level of V_E can be calculated from.

$$V_E = V_B - V_{BE} \rightarrow (3)$$

and the emitter current can be determined from

$$I_E = \frac{V_E}{R_E} \rightarrow (4)$$

$$I_C \equiv I_E \rightarrow (5)$$

The Collector to emitter Voltage is determined by

$$V_{CE} = V_{CC} - I_C R_C - I_E R_E$$

$$V_{CE} = V_{CC} - I_C (R_C + R_E) \rightarrow (6)$$

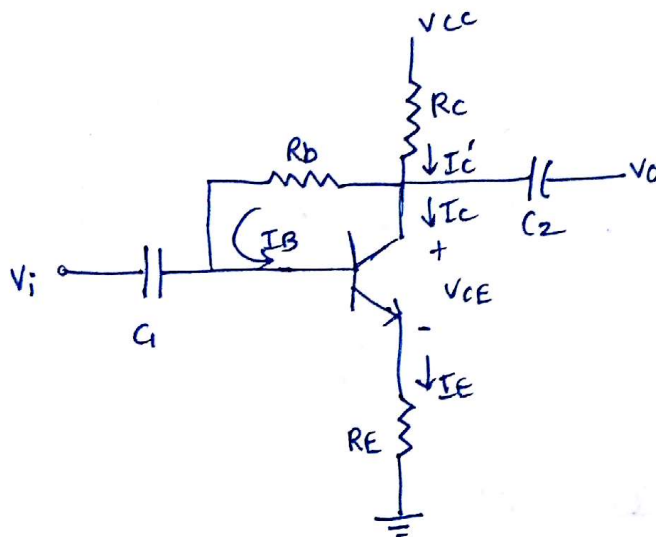
In this approximation I_B was not calculated.

The Q point by I_{CQ} and V_{CEQ} is therefore independent of value of β

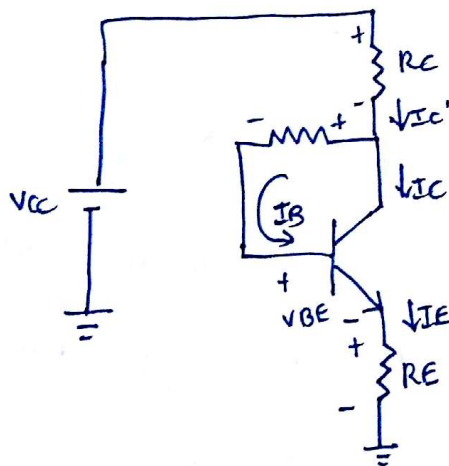
DC Bias with Voltage feedback

An improved level of stability can also be obtained by introducing a feedback path from collector to base as shown in figure.

Although the Q point is not totally independent of β , the sensitivity to changes in β or temperature variations is normally less than encountered for the fixed bias or emitter bias configuration.



Base Emitter loop



It is important to note that the current through R_c is not I_c but I_c' (where $I_c' = I_c + I_B$)

By the approximation $I_c' \equiv I_c$ is normally employed.

Substituting $I_c' \equiv I_c \equiv \beta I_B$ and $I_E \equiv I_c$

$$V_{CC} - \beta I_B R_c - I_B R_B - V_{BE} - \beta I_B R_E = 0$$

Gathering the terms.

$$V_{CC} - V_{BE} - \beta I_B (R_c + R_E) - I_B R_B = 0$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta(R_c + R_E)}$$

Since $I_c = \beta I_B$.

$$I_c = \frac{\beta (V_{CC} - V_{BE})}{R_B + \beta(R_c + R_E)}$$

Collector emitter loop

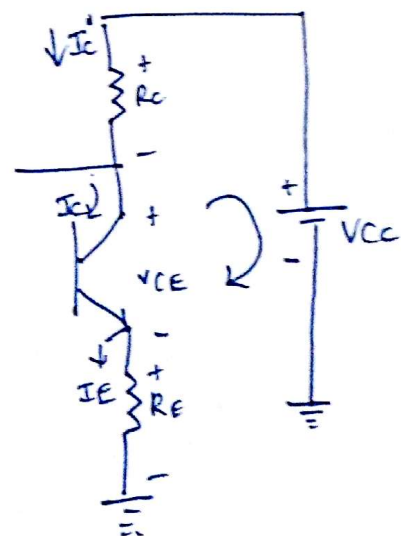
Applying KVL

$$I_E R_E + V_{CE} + I_c' R_c = V_{CC}$$

Since $I_c' \equiv I_c$ & $I_E \equiv I_c$

$$I_c (R_c + R_E) + V_{CE} - V_{CC} = 0$$

$$V_{CE} = V_{CC} - I_c (R_c + R_E)$$



FET Biasing

In field effect transistor, the relationship between input and output quantities is non linear due to squared term in Shockley equation.

Linear relationship result in straight lines when plotted on a graph of one variable versus the other, while nonlinear due to the function results in curves as obtained for the transfer characteristics of a JFET

The non linear relationship between I_D and V_{GS} can complicate the mathematical approach to the dc analysis of FET configuration.

Another distinct difference between the analysis of BJT and FET transistor is that the input controlling variable for a BJT transistor is current level, while for the FET a voltage is the controlling variable.

The general relationship that can be applied to the dc analysis of all FET amplifiers are

$$I_{G1} \approx 0 \text{ A}$$

$$I_D = I_S$$

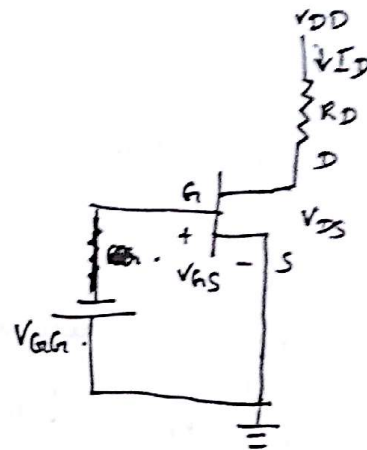
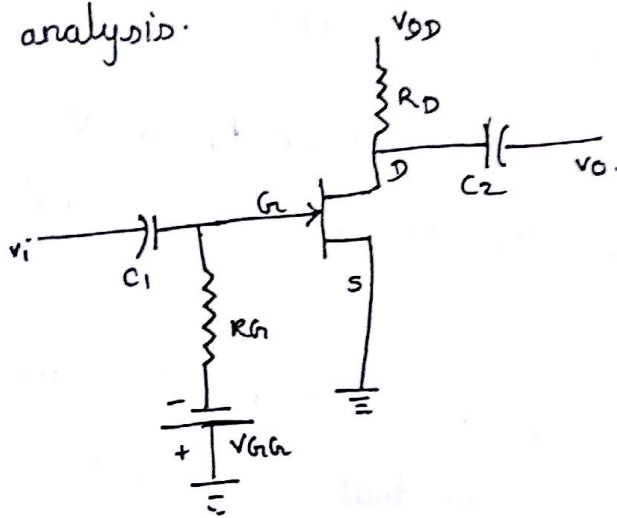
For JFET and depletion type MOSFET, Shockley's equation is applied to relate the input and output quantities.

$$I_D = I_{DSS} \left(\frac{1 - V_{GS}}{V_P} \right)^2$$

Fixed Bias Configuration.

The simplest of biasing arrangements for the n-channel JFET shown in figure.

The configuration includes the ac levels V_i and V_o and coupling capacitors (C_1 and C_2). The coupling capacitors are open circuit for dc analysis and low impedance for ac analysis.



Fixed bias Configuration.

For DC analysis.

$$I_G \equiv 0A$$

$$V_{RG} = I_G R_G$$

$$= (0A) R_G$$

$$= 0V$$

The zero volt drop across R_G permits replacing R_G by a short circuit equivalent

The fact that the negative terminal of the battery is connected directly to the defined positive potential of V_{GS} clearly reveals that the polarity of V_{GS} is directly opposite to that of V_{GG} .

Applying KVL in clock wise direction.

$$-V_{GG} - V_{GS} = 0$$

$$V_{GS} = -V_{GG}$$

Since V_{GG} is fixed in dc supply and V_{GS} is fixed in magnitude.

The resulting level of drain current I_D is now controlled by shockley equation

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

The drain to source voltage of the output section can be determined by applying KVL

$$V_{DS} + I_D R_D - V_{DD} = 0$$

$$V_{DS} = V_{DD} - I_D R_D$$

$$V_S = 0V$$

using double subscript notation

$$V_{DS} = V_D - V_S$$

$$V_D = V_{DS} + V_S$$

$$V_D = V_{DS}$$

$$V_{GS} = V_G - V_S$$

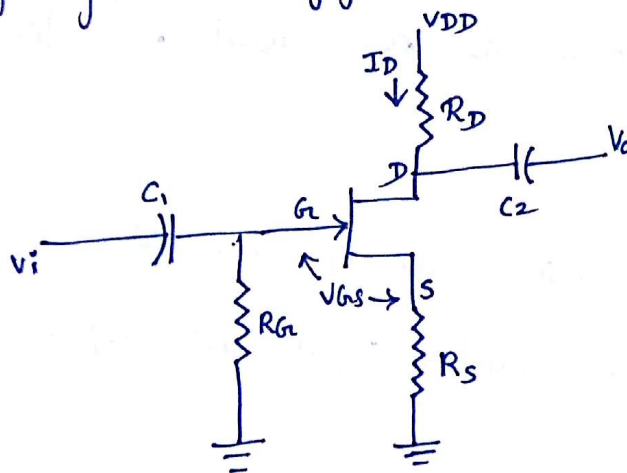
$$V_{GS} = V_{GS} + V_S$$

$$V_G = V_{GS}$$

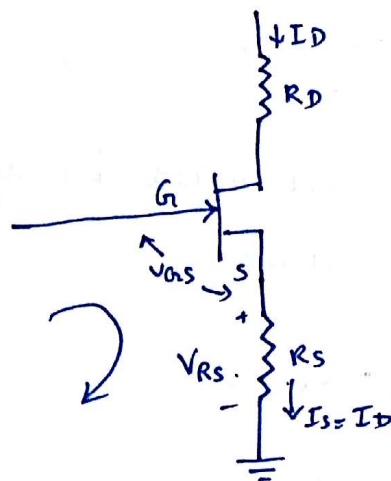
Self bias Configuration.

The self bias Configuration eliminates the need for two dc supplies.

The Controlling gate to Source voltage is now determined by the voltage across a resistor R_S introduced in the source leg of the configuration.



For DC analysis the Capacitors can again be replaced by open circuit. and the resistor R_G replaced by a short circuit equivalent since $I_G = 0A$.



The current through R_S is the source current I_S but $I_S = I_D$.

~~For the~~

$$V_{RS} = I_D R_S$$

For the indicated closed loop of figure

$$-V_{GS} = V_{RS}$$

$$-V_{GS} - V_{RS} = 0$$

$$V_{GS} = -V_{RS}$$

$$V_{GS} = -I_D R_S \rightarrow \textcircled{1}$$

Note in this case that V_{GS} is function of output current I_D and not fixed in magnitude as occurred for fixed bias configuration

Equation $\textcircled{1}$ is defined by the network configuration and shockley equation relates input and output quantities.

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

$$= I_{DSS} \left(1 - \frac{-I_D R_S}{V_P} \right)^2$$

$$I_D = I_{DSS} \left(1 + \frac{I_D R_S}{V_P} \right)^2$$

The level of V_{DS} can be determined by applying Kirchhoff's voltage law to the output circuit

$$V_{RS} + V_{DS} + V_{RD} - V_{DD} = 0$$

$$V_{DS} = V_{DD} - V_{RS} - V_{RD}$$

$$V_{DS} = V_{DD} - I_S R_S - I_D R_D$$

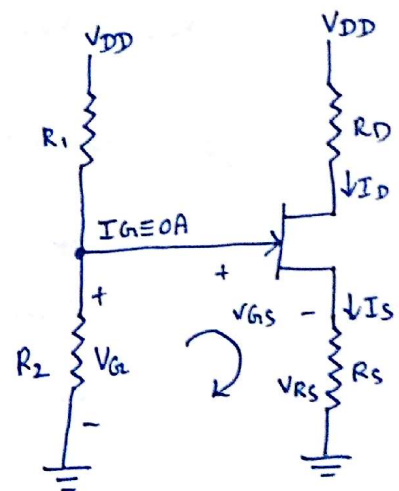
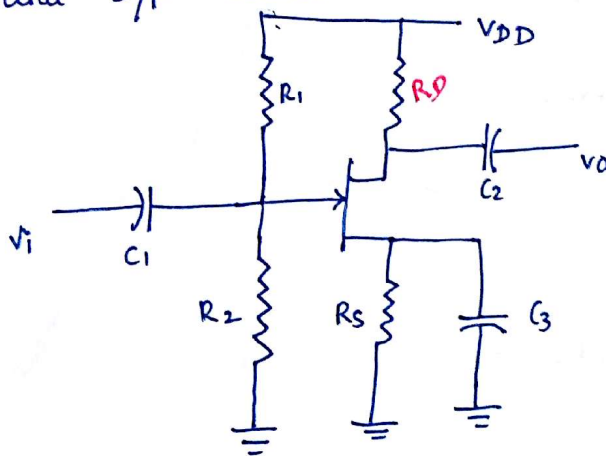
$$I_D = I_S$$

$$V_{DS} = V_{DD} - I_D (R_S + R_D)$$

Voltage Divider Biasing

The voltage divider Bias arrangement applied to BJT transistor amplifier is also applied to FET amplifiers.

The Basic construction is exactly the same but dc analysis of each is quite different. $I_G = 0A$ for FET amplifiers but the magnitude of I_B for common emitter BJT amplifiers can affect the dc levels. of current and voltage in both i/p and o/p circuits.



The Source V_{DD} was separated into two equivalent source to permit a further separation of input and output regions of network.

Since $I_G = 0A$ KCL requires that $I_{R_1} = I_{R_2}$

The voltage V_G equal to the voltage across R_2 can be found using voltage divider rule.

$$V_G = \frac{V_{DD} R_2}{R_1 + R_2}$$

Applying Kirchhoff's voltage law in clockwise direction to the indicated loop.

$$V_{G2} - V_{GS} - V_{RS} = 0$$

$$V_{GS} = V_{G2} - V_{RS}$$

Substituting $V_{RS} = I_S R_S = I_D R_S$

$$V_{GS} = V_{G2} - I_D R_S$$

Apply KVL to o/p side

$$V_{RS} + V_{DS} + V_{RD} - V_{DD} = 0$$

$$V_{DS} = V_{DD} - V_{RS} - V_{RD}$$

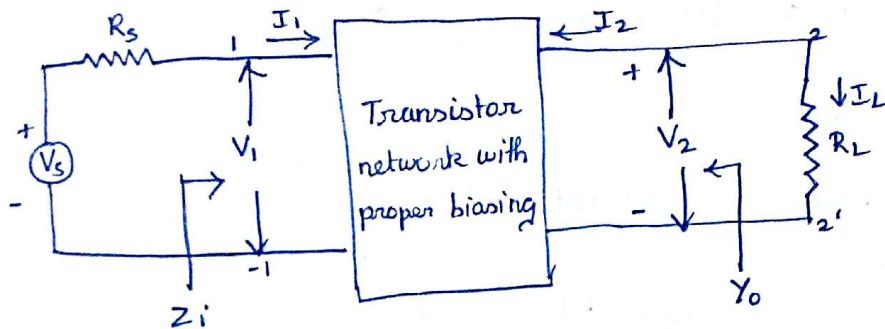
$$V_{DS} = V_{DD} - I_S R_S - I_D R_D$$

$$I_D = I_S$$

$$V_{DS} = V_{DD} - I_D (R_S + R_D)$$

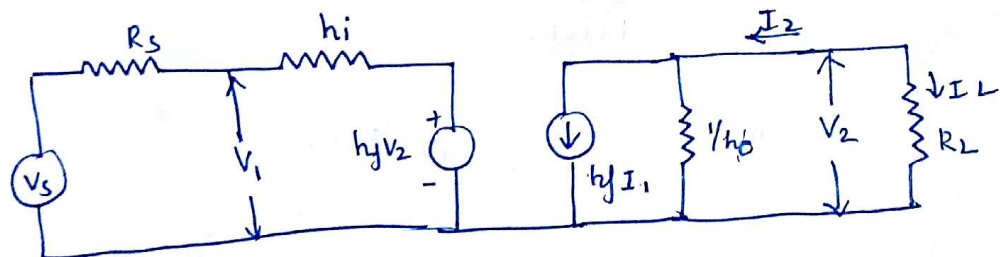
Transistor Modeling

Figure shows the basic transistor amplifier only it is necessary to connect an external load and signal source along with Proper biasing.



Basic transistor amplifier

Replace transmission circuit with its small hybrid model as shown in figure.



Important Parameter of BJT

Let us analyze hybrid model to find the current gain, the input resistance, the voltage gain and output resistance.

Current gain

For transistor amplifier (A_i) is defined as the ratio of output to input current

$$A_i = I_o / I_i$$

$$A_i = \frac{-I_2}{I_1} \rightarrow (1)$$

$\rightarrow I_1$ & I_2 are equal in magnitude but opposite in sign (ie)
 $I_1 = -I_2$

$\rightarrow$ from the circuit

$$I_2 = h_f I_1 + h_o V_2 \rightarrow (2)$$

$$V_2 = -I_2 R_1$$

Substitute $V_2 = -I_2 R_1$ in equation (2)

$$I_2 = h_f I_1 + h_o (-I_2 R_1)$$

$$I_2 + h_o I_2 R_1 = h_f I_1$$

$$I_2 (1 + h_o R_1) = h_f I_1$$

$$\frac{I_2}{I_1} = \frac{h_f}{1 + h_o R_1}$$

$$A_i = \frac{-I_2}{I_1} = \frac{-h_f}{1 + h_o R_1}$$

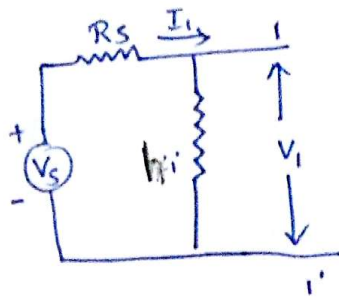
Current gain (A_{I_s})

It is the current gain taking into account for the source resistance R_s , if the model driven by current source instead of voltage source. It is given by

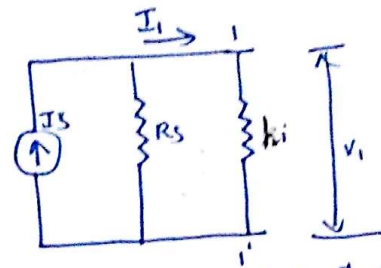
$$A_{I_s} = \frac{-I_2}{I_s}$$

$$= \frac{-I_2}{I_1} \cdot \frac{I_1}{I_s}$$

$$= A_i \cdot \frac{I_1}{I_s}$$



Input section of hybrid model.



Input section of hybrid model with current source instead of voltage source.

Apply current divider rule for the figure

$$I_1 = \frac{I_s R_s}{h_i + R_s}$$

$$A_{I_s} = \frac{I_1}{I_s} \times A_i$$

$$= A_i R_s / h_i + R_s$$

Input Impedance (Z_i)

From the figure, R_i is the input resistance looking into the amplifier. input terminal (1, 1') is given by

$$Z_i = \frac{V_1}{I_1}$$

For input circuit

$$V_1 = h_i I_1 + h_r V_2$$

$$Z_i = \frac{V_1}{I_1}$$

$$= \frac{h_i I_1 + h_r V_2}{I_1}$$

$$= h_i + \frac{h_r V_2}{I_1}$$

Substitute $V_2 = -I_2 R_1 = A_i R_1 I_1$ then.

$$Z_i = h_i + \frac{h_{r2} A_i I_1 R_1}{I_1}$$

$$= h_i + h_{r2} A_i R_1$$

Substitute $A_i = \frac{-h_f}{1} + h_o R_1$

$$Z_i = h_i - h_{r2} h_f R_1 / 1 + h_o R_1$$

So Z_i is the input impedance of the common emitter circuit.

Voltage gain (A_v)

It is the ratio of o/p voltage to input voltage.

$$A_v = \frac{V_2}{V_1}$$

$$A_v = \frac{A_i I_1 R_1}{V_1}$$

Voltage Gain (A_{vs})

→ It is voltage gain including the source.

$$A_{vs} = \frac{V_2}{V_s}$$

$$= \frac{V_2}{V_1} * \frac{V_1}{V_s}$$

$$= A_v * \frac{V_1}{V_s}$$

→ From figure (a) apply voltage divider rule.

$$V_1 = \frac{V_s \cdot h_i}{h_i + R_s}$$

$$\frac{V_i}{V_s} = \frac{h_i}{h_i + R_s}$$

Substitute $\frac{V_i}{V_s}$ in A_{vs}

$$A_{vs} = A_v \times \frac{h_i}{h_i + R_s}$$

Output Admittance (Y_o)

It is the ratio of output current I_2 to output voltage V_2

$$Y_o = \frac{I_2}{V_2} \quad \text{with } V_1 = 0$$

W.K.T

$$I_2 = h_f I_1 + h_o V_2$$

$$\frac{I_2}{V_2} = \frac{h_f I_1}{V_2} + h_o$$

$$Y_o = \frac{h_f I_1}{V_2} + h_o$$

Power gain (A_p)

It is the ratio of average power delivered to the load R_L to the output power.

$$P_2 = V_2 I_L$$

$$= -V_2 I_2$$

Since the input power $P_1 = V_1 I_1$, the operating power gain A_p of

the transistor

$$A_p = \frac{P_2}{P_1} = \frac{-V_2 I_2}{V_1 I_1} = A_v A_i$$

Relation between A_{Vs} & A_{Is}

$$A_{Vs} = \frac{A_i R_L}{Z_i} + R_s$$

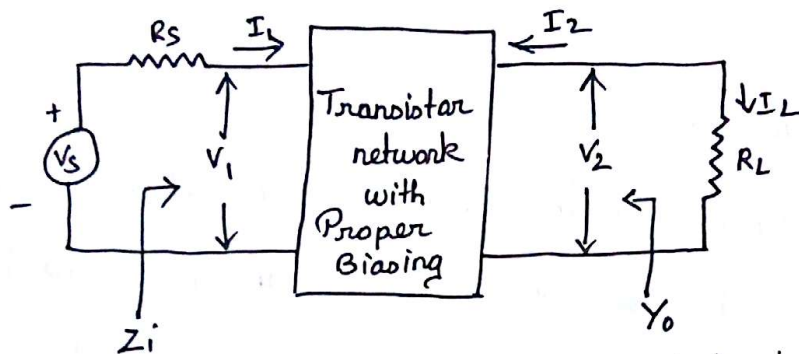
$$A_{Is} = \frac{A_i R_s}{Z_i} + R_s$$

$$\frac{A_{Vs}}{A_{Is}} = \frac{R_L}{R_s}$$

$$A_{Vs} = A_{Is} \frac{R_L}{R_s}$$

H Parameter Model.

* Let us consider transistor amplifier as shown in figure.



* The transistor is a current operated device ; input current i is an independent variable.

The input current and output voltage devices drives the input voltage as well as output current.

Thus the equation can be written as

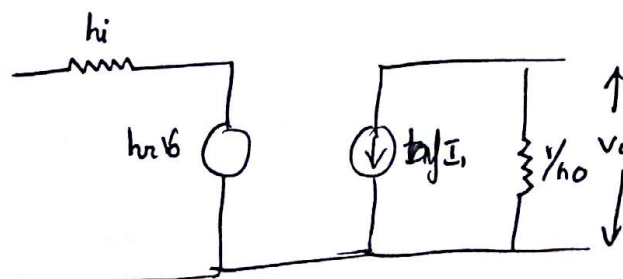
$$V_i = h_{11}I_i + h_{12}V_o$$

$$I_o = h_{21}I_i + h_{22}V_o$$

It also written in alphabet notation.

$$V_i = h_i I_i + h_r V_o$$

$$I_o = h_f I_i + h_o V_o$$



a) With output short circuited

$H_{11} = h_i$ - input resistance.

$H_{21} = h_f$ - short circuited current gain.

b) With Input open circuited

$H_{12} = h_{r}$ - Reverse voltage transfer ratio.

$H_{22} = h_o$ - output admittance .

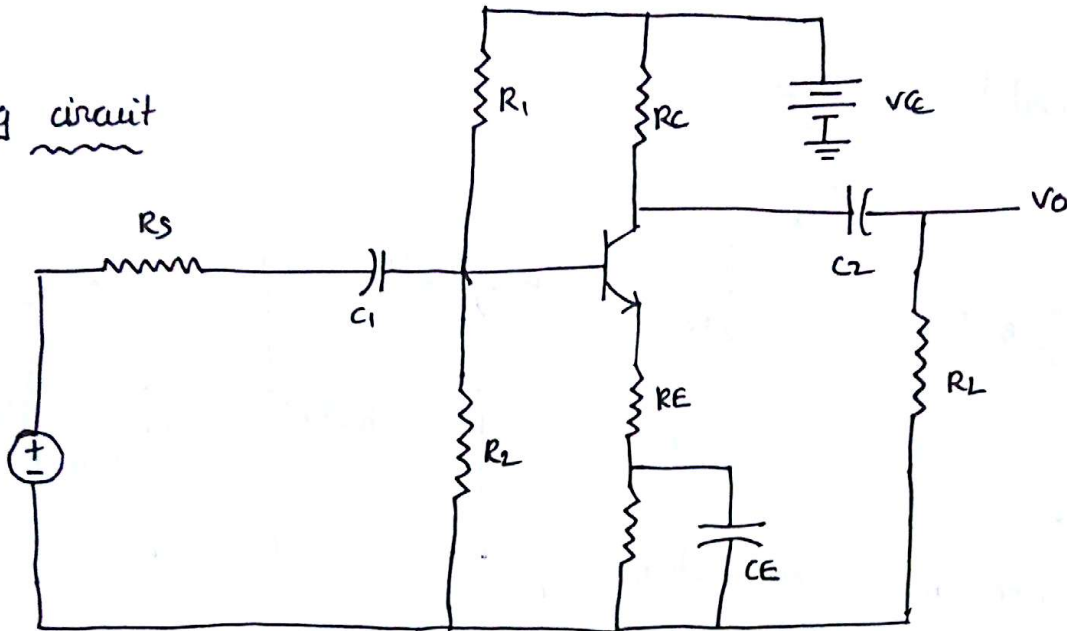
Benefits

- * Easy to measure
- * Can be obtained from the transistor characteristic curve
- * Convenient to use in circuit analysis and design.
- * Most of the transistor manufacturers specify the h-parameter

Common emitter amplifier

* Figure shows the practical circuit of common emitter transistor amplifier. It consists of differential circuit components are

Biassing circuit



Input Capacitance

It couples the signal to the base of transistor. It blocks any DC component present in the signal and passes only AC signal for amplification.

Emitter Bypass Capacitance

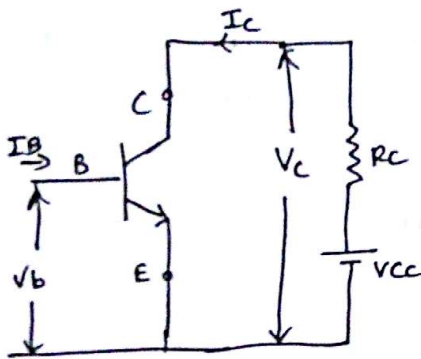
It is connected in parallel with the emitter resistance R_E to provide low reactance path to the amplified AC signal.

The amplified AC signal passed through R_E will cause voltage drop across it. This will reduce the output.

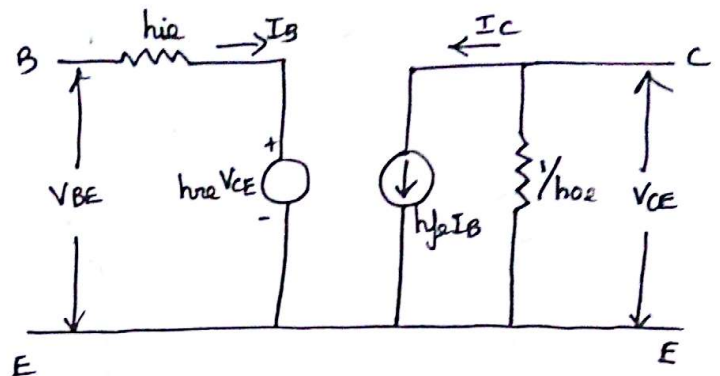
output Coupling Capacitor

It couples the output of the amplifier to the load or to next stage of amplifier. It blocks DC and passes only AC part of amplifier signal.

Hybrid model for CE



Common emitter configuration.



From the h-Parameter model

$$V_{be} = h_{ie} I_b + h_{re} V_{ce}$$

$$I_c = h_{fe} I_b + h_{oe} V_{ce}$$

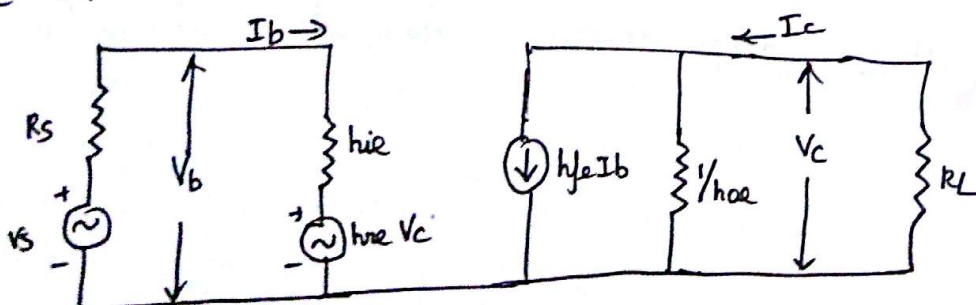
$$\text{Where } h_{ie} = \left. \frac{\Delta V_{be}}{\Delta I_b} \right|_{V_{ce} \text{ constant}}$$

$$h_{re} = \left. \frac{\Delta V_{be}}{\Delta V_{ce}} \right|_{I_b \text{ constant}}$$

$$h_{fe} = \left. \frac{\Delta I_c}{\Delta I_b} \right|_{V_{ce} \text{ constant}}$$

$$h_{oe} = \left. \frac{\Delta I_c}{\Delta V_{ce}} \right|_{I_b \text{ constant}}$$

Analysis of CE amplifier



Reg No _____
Since $1/h_{oe}$ is in parallel with R_L if $1/h_{oe} \gg R_L$, then h_{oe} may be neglected

Current gain.

$$A_i = \frac{-I_c}{I_b} \rightarrow (1)$$

$$I_c = h_{fe} I_b + \frac{1}{h_{oe}}$$

$$~~I_c = h_{fe} I_b +~~$$

substitute in equation 1

$$= \frac{-h_{fe} I_b + 1/h_{oe}}{I_b}$$

By neglecting h_{oe}

$$A_i = -h_{fe}$$

Input Impedance

$$Z_i = h_{ie} + h_{re} A_i R_L$$

By neglecting h_{re}

$$Z_i = h_{ie}$$

voltage gain.

$$A_v = \frac{A_i R_L}{Z_i} = \frac{A_i R_L}{h_{ie}}$$

output admittance

$$Y_o = \frac{h_{oe} - h_{fe} h_{re}}{h_{ie} + R_s}$$

By neglecting h_{oe} & h_{re}

$$Y_o = 0$$

Important Parameters of JFET

* The drain to source current of JFET is controlled by gate to source voltage.

* The change in drain current due to change in gate to source voltage can be determined using the transconductance factor.

$$\Delta I_d = g_m \Delta V_{gs}$$

* In BJT the relation between an output and input quantity is given by amplification factor β , where in JFET the relation is given by g_m .

* The another important parameter of JFET is drain Resistance R_d .

$$R_d = \left. \frac{\Delta V_{ds}}{\Delta I_{ds}} \right|_{V_{gs} \text{ constant}}$$

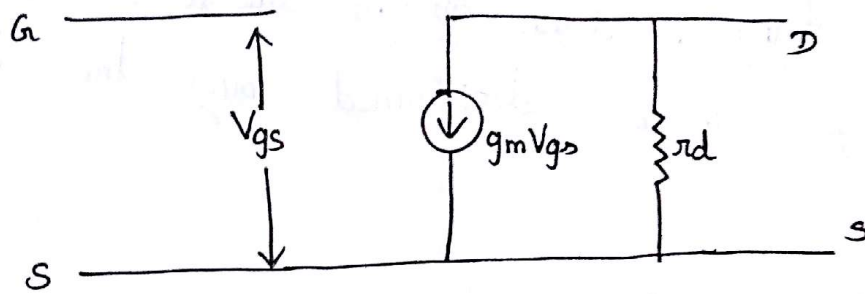
* It determines the output admittance z_o of FET amplifier.

Small signal model and Parameters

Figure shows the small signal low frequency AC equivalent circuit for n channel JFET.

The input impedance is represented by the open circuit at its input terminals, since gate current I_G is 0.

The output impedance by r_d from drain to source

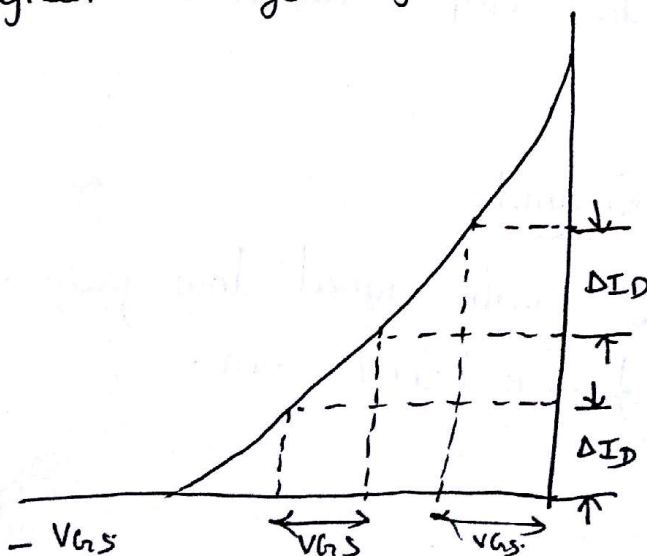


Parameters

- * Transconductance (g_m)
- * Input Resistance & capacitance.
- * Drain Source Resistance (r_d)
- * Amplification factor (μ)

Transconductance

The transconductance g_m is the change in drain current for a given change in gate to source voltage.



$$\frac{\Delta I_D}{\Delta V_{GS}} \Big|_{V_{DS} \text{ constant}}$$

From figure we noted that slope varies, g_m also varies.

$$g_m = \frac{\Delta I_d}{\Delta V_{GS}} \quad \left| \quad V_{DS} \text{ constant} \right.$$

For given g_m , we can calculate an approximate value for g_m at any point on transfer characteristic of curve

$$g_m = g_{m0} [1 - V_{GS}/V_{GS}(\text{off})]$$

Where g_{m0} is the value of g_m for $V_{GS} = 0$

Input Resistance and Capacitance

A JFET operates with its gate source junction reverse biased. Therefore the input resistance at gate is very high.

$$R_{IN} = |V_{GS}/I_{GS}|$$

The input resistance decreases with increase in temperature.

Input capacitance is based on ~~to~~ how much reverse voltage applied.

Drain to Source Resistance

Drain resistance r_d is the AC resistance between drain and source terminals upon the JFET are operating in the saturation Region.

$$R_d = \frac{\Delta V_{DS}}{\Delta I_D}$$

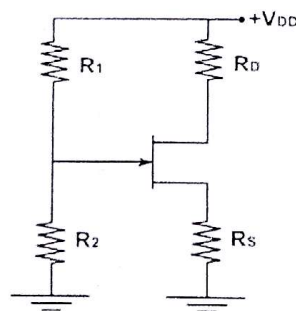
2.8 MOSFET:

- To analyze and design the dc biasing of MOS transistor circuits. As with bipolar, the dc biasing of MOSFETs, is an importance part of the design of amplifiers. Here, resistors are used in junction with the MOS transistor. In real MOSFET integrated circuit, however, the resistors are generally replaced by other MOSFETs, so the circuit is composed entirely of MOS devices. In the dc analysis of MOSFET circuits, we can use the ideal current-voltage equations listed in table.

Si.no	nMOS	pMOS
1	Non saturation region $V_{ds} < V_{ds(sat)}$ $I_d = k[2(V_{gs} - V_t)V_{ds} - V_{ds}^2]$	Non saturation region $V_{sd} < V_{sd(sat)}$ $I_d = k[2(V_{sg} - V_t)V_{sd} - V_{sd}^2]$
2	Saturation region $V_{DS} > V_{DS(sat)}$ $I_p = k(V_{GS} - V_T)^2$	Saturation region $V_{SD} > V_{SD(sat)}$ $I_p = k(V_{SG} + V_T)^2$
3	Transition point $V_{DS(sat)} = V_{GS} - V_T$	Transition point $V_{SD(sat)} = (V_{SG} - V_T)$
4	Enhancement mode $V_T > 0$	Enhancement mode $V_T < 0$
5	Depletion mode $V_T < 0$	Depletion mode $V_T > 0$

2.8.1 COMMON SOURCE CIRCUIT FOR EMOSFET WITH SOURCE RESISTOR

- Voltage divider bias circuit for W- Channel enhancement type MOSFET is shown in fig.



As $I_G = 0$.

$$V_G = R_2 V_{DD} / (R_1 + R_2)$$

Applying KVL to the input circuit, we get

$$V_G - V_{GS} - V_S = 0$$

$$V_{GS} = V_G - I_S R_S = V_G - I_D R_D$$

$$V_{GS} = V_G - I_D R_S$$

Applying KVL to the output circuit, we get

$$V_{DD} - I_D R_D - V_{DS} - I_S R_S = 0$$

$$V_{DS} = V_{DD} - I_D R_D - I_S R_S$$

$$V_{DS} = V_{DD} - I_D (R_D + R_S)$$

Biasing Circuit for D – MOSFET:

- Biasing Circuit for depletion type MOSFET are quite similar to the circuits used for FET biasing
- The Primary difference between the two is the fact that depletion type MOSFET's also permit operating points with positive value of V_{GS} for n-channel and negative values of V_{GS} for p-channel MOSFET.
- To have positive value of V_{GS} for n-channel and negative value of V_{GS} for p-channel self-bias circuit is unsuitable.